

**A Neuro-Inspired Psychometric Assessment System for Workplace
Crisis Resilience**

PHASE I REPORT

Submitted by

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ANNA UNIVERSITY, CHENNAI
BONAFIDE CERTIFICATE

Certified that this Report titled “PSYCOLOGICAL EVALUATION SYSTEM”: A NEURO-INSPIRED PSYCHOMETRIC ASSESSMENT SYSTEM FOR WORKPLACE CRISIS RESILIENCE" is the bonafide work of AKSHITHAA H (230701025) and FARHEEN TABASSUM H (230701088) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

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Submitted to Project Viva-Voce Examination held on _____

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ABSTRACT

Both behavioral interviewing and workplace stress evaluations are very subjective exercises, since the judgement of two separate panels reading the same response could differ. An interviewee who prepares their answers ahead of time may look more resilient than he actually is. PSYCOLOGICAL EVALUATION SYSTEM is a working prototype that aims to make such judgement more objective and explainable. In PSYCOLOGICAL EVALUATION SYSTEM a candidate is presented with a realistic work crisis in a web interface and asked to give an answer. Their response is analyzed semantically by a Gemma-based model running on Google Colab or a local rule-based analyzer in case Google Colab is not accessible. As a result of analysis we get six psychometric indicators derived from Perceived Stress Scale, State-Trait Anxiety Inventory and a salience-style inventory, and those indicators are converted into seven values corresponding to the activity of seven Yeo networks. Those values are further combined by multi-layer perceptron in a composite resilience score, which is benchmarked against scenarios provided in a CSV file and transformed into plain-language statements regarding the candidate's ability to handle workplace stress, burnout proneness, emotional shifts and adaptability. It is a notable feature that our system does not analyze or measure any individual's brain.

The seven values of network activity are interpretative estimates calibrated against the target values provided along with the project, and all outputs of the system are marked as non-clinical prototype results. Contribution is thus made on the level of system design: a clear trace from text through indicators, network estimates and score calculation steps can be checked, there is a guaranteed valid response in case of failure mode, and an interface made for review by human panel. On 6 test responses the MLP reached $R^2 = 0.976$ against its calibration targets, and the end-to-end pipeline completed in a median of 1.18 seconds. Criterion validity against real job performance, fairness testing and validation with neuroimaging are left as future work.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
BOLD	Blood-Oxygen-Level-Dependent
CORS	Cross-Origin Resource Sharing
CSV	Comma-Separated Values
DMN	Default Mode Network
fMRI	Functional Magnetic Resonance Imaging
HITL	Human-in-the-Loop
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
LLM	Large Language Model
MAE	Mean Absolute Error
MLP	Multi-Layer Perceptron
PSS	Perceived Stress Scale
REST	Representational State Transfer
RQ	Research Question
SRSI	Stress Response and Saliency Inventory
STAI	State-Trait Anxiety Inventory
UI	User Interface
Yeo 7	Seven-network cortical parcellation of Yeo et al.

CHAPTER 1 INTRODUCTION

1.1 GENERAL

Recruiters and managers face a similar dilemma: the candidate who does well in the interview is not necessarily the one who handles problems well. A relaxed and reflective individual might appear to be reserved in the interview process, whereas an outgoing individual might feel frozen during the first situation he encounters where things go bad. The behavioral interview is a tool that helps predict behavior during stress, but the result of that interview is highly dependent on the listener's fatigue level and persuasiveness of the candidate [1]. Structured interviews reduce this variance, although they don't eliminate it [2].

Stress tolerance is an especially difficult quality to judge from a conversation. Psychology offers validated questionnaires such as the Perceived Stress Scale [3] and the State-Trait Anxiety Inventory [4], but they ask people to describe themselves in general terms. They say little about how a person actually reasons through a specific crisis. A written answer to a concrete scenario is a richer signal, because the way a person structures a response, whom they involve, how they sequence actions and how they talk about uncertainty all carry information.

This project turns such a written answer into something a panel can inspect. The answer is analysed into six interpretable psychometric indicators, expressed through the seven large-scale cortical networks of the Yeo parcellation [5], passed through a multi-layer perceptron, and summarised as a resilience score with plain-language explanations. The network vocabulary is borrowed from network neuroscience, where executive and attention networks are linked to regulating a response and limbic, somatomotor and default-mode activity are linked to arousal and rumination [6], [7].

It is important to be precise about what the system does. It does not scan, record or measure anyone's brain, and the seven network values are estimates produced by software. They give the score a biologically motivated structure that makes it easier to explain, but they are not neuroimaging results. The prototype also states this on screen: every output is marked as a non-clinical, demonstration-level result.

1.2 OBJECTIVE

The objective of this phase is to design and build a working end-to-end prototype with the following goals:

- Present a scenario-based interview question in a web interface, one question at a time, and accept a free-text answer from the candidate.
- Analyse the answer semantically using a Gemma-based model hosted on Google Colab, and convert the result into seven attributes corresponding to the Yeo cortical networks.
- Score the seven attributes with a multi-layer perceptron and compare the result against benchmark scenario data held in a CSV file.
- Generate readable behavioural interpretations covering stress handling, burnout risk, emotional shifts and adaptability, together with a visible audit trail of every step.
- Guarantee that a valid result is always returned, by falling back automatically to a local deterministic analyser when the hosted model is unavailable, without exposing raw errors to the user.
- Keep a human decision-maker in control, and mark every result clearly as a non-clinical prototype output.

1.3 EXISTING SYSTEM

In most organisations, resilience and stress tolerance are assessed through a mix of unstructured interview questions, self-report questionnaires and the interviewer's impression. Each of these has a weakness. Interview impressions vary between interviewers, questionnaires can be answered strategically, and neither explains why a particular candidate received a particular rating.

Such tools are available, but they are quite limited in scope. The sentiment classifier will tag a text as positive or negative – a very bad indicator of coping strategy. The commercial systems for video analysis claim that they can evaluate the personality through face and voice and they got quite a lot of criticism for insufficient evidence and

bias. All these tools provide a single result with no explanation and that is why it is difficult to challenge the outcome by a panel of people. It is also worth noting that the existing approaches do not provide the place of human being in the system. The score is either completely ignored or believed and the intermediate logic cannot be questioned.

1.4 PROPOSED SYSTEM

PSYCHOLOGICAL EVALUATION SYSTEM proposes an open pipeline from the answer to the resilience profile. A Flask backend receives the candidate's answer, sends it to a Gemma semantic analysing pipeline running on Google Colab, gets six psychometric indicators, which are mapped to seven network properties, evaluated using a multi-layer perceptron, compared against benchmarking cases in `tribev2_output.csv` and turned into narrative interpretations.

In case Google Colab is not available, the same pipeline is executed on the local machine with a deterministic analyser, so the front-end will always receive a valid result. The web interface displays the status of each step on the console, final score, seven bars with network values and four interpretive cards. The server-side console receives the identical output, but in a banner form. The main idea throughout the entire architecture is that nothing important should hide behind a single number.

CHAPTER 2 LITERATURE SURVEY

This chapter reviews the work that informs the project, identifies what is missing from it, states the contribution of this work, and lists the research questions that guide the evaluation.

2.1 LITERATURE REVIEW

Interview validity. Levashina et al. [1] reviewed the structured employment interview and showed that standardised questions and explicit scoring rules improve consistency between interviewers. Schmidt and Hunter [2] summarised evidence from many decades of selection research and highlighted how limited the predictive power of individual methods can be. These studies support the decision to use a fixed scenario with an explicit scoring scheme rather than an open conversation.

Stress and anxiety measurement. The Perceived Stress Scale of Cohen et al. [3] measures how unpredictable, uncontrollable and overloaded a person judges their life, while Spielberger's State-Trait Anxiety Inventory [4] distinguishes momentary anxiety from a stable tendency. The six indicators used here borrow the constructs these instruments measure, such as perceived control, perceived threat and arousal, and apply them to text rather than to questionnaire items.

Large-scale brain networks. Yeo et al. [5] derived seven cortical networks from resting-state connectivity in one thousand participants, and the parcellation has become a common reference frame. Menon [6] proposed that psychiatric conditions can be understood through interactions among a small number of large-scale networks, and Arnsten [7] described how stress signalling weakens prefrontal regulation. Raichle et al. [8] characterised the default mode of brain function, which is often linked to self-referential thought and rumination. These findings provide a principled way to group psychological indicators under network headings, but they do not allow an individual's brain activity to be predicted from an essay.

Language representation. Sentence-level embedding models such as Sentence-BERT [9] make it practical to convert short free text into numeric vectors for downstream models. Brain-encoding foundation models such as Meta's TRIBE v2 [10] predict cortical responses to video, audio and language stimuli, but they are trained on people who were scanned while experiencing those stimuli and they predict population-level responses. This project borrows the idea of describing activity through Yeo networks but does not reproduce or extend such a model.

Human oversight. Adedokun et al. [11] reviewed human-in-the-loop frameworks, oversight mechanisms and trust calibration for agentic AI, and Cheng and Cheng [12] proposed a decoupled architecture for controlled human intervention. Both support designing the assessment output as input to a person rather than as an automated verdict.

2.2 COMPARISON AND DISCUSSION

Table 2.1 compares the strands of related work with the proposed system. Interview research addresses reliability, questionnaire research addresses measurement, neuroscience

addresses network organisation, and oversight research addresses accountability, but each stops at its own boundary.

Table 2.1. Comparison with related approaches

Approach	Free text	Psycho-metric	Network view	Explain-able	HITL
Structured interviews [1], [2]	–	–	–	–	Yes
PSS / STAI [3], [4]	–	Yes	–	–	–
Brain-encoding models [10]	Yes	–	Yes	–	–
HITL frameworks [11], [12]	–	–	–	–	Yes
This work	Yes	Yes	Yes*	Yes	Yes

* Interpretive network estimates, not measured neural activity.

2.2.1 RESEARCH GAP

The cited study has created a practical lacuna. There is no observable pipeline which takes as input the candidate’s language-based response to a crisis situation and outputs a set of indicators based on psychological theories, represented in a network terminology format which facilitates explanation, compared with reference cases, and presented to the human panelists.

2.2.2 PROPOSED CONTRIBUTION / NOVELTY

The contribution of this project lies in system- and workflow-level integration rather than in a new learning algorithm. Specifically: (i) a transparent text-to-indicator-to-network-to-score chain in which every intermediate value is displayed; (ii) a dual-mode execution design in which a hosted Gemma model is used when available and a deterministic local analyser guarantees a result otherwise; and (iii) a candidate-facing interface that gives no hints and an evaluator-facing console that shows the full audit trail, with all outputs labelled as non-clinical.

2.3 RESEARCH QUESTIONS

Four research questions guide the work.

- RQ1: Can a free-text response to a crisis scenario be projected onto a small set of interpretable psychometric indicators consistently?
- RQ2: Is there a regression model capable of mapping those indicators to estimates of Yeo networks and reproducing its calibration targets?
- RQ3: Is there a dual-mode approach capable of ensuring availability and validity of output even when the hosted model becomes unavailable?
- RQ4: Is there a comparison with benchmarking cases and an interface capable of allowing the human user to make their decision?

2.4 CONCLUSION

These building blocks have separate literature support: standardised scenarios, valid stress concepts, common network language, textual embeddings, and human in-the-loop supervision. There is no literature on combining them into an integrated, transparent, and truthful application. The following chapters discuss how this project integrates them.

CHAPTER 3 SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The prototype employs a layered approach and consists of three main components. First, there is the browser-based interface (index.html), which provides the scenario, accepts the response, and opens the live console. Second, there is the Flask backend (server.py) exposing the scenario and submission API endpoints, which controls the llm.py pipeline implementation. The third is an optional hosted analysis service (colab_server.py) that runs on Google Colab, performs the Gemma-based semantic analysis and the MLP prediction, and is reached through a public tunnel URL.

When a candidate submits an answer, the backend first attempts to call the Colab service. If the call succeeds, the returned indicators and prediction are used. If the call fails, times out or returns an error, the backend runs the same stages locally with deterministic logic. In both cases the result is compared with the benchmark file, turned into interpretations and returned to the interface and to the terminal.

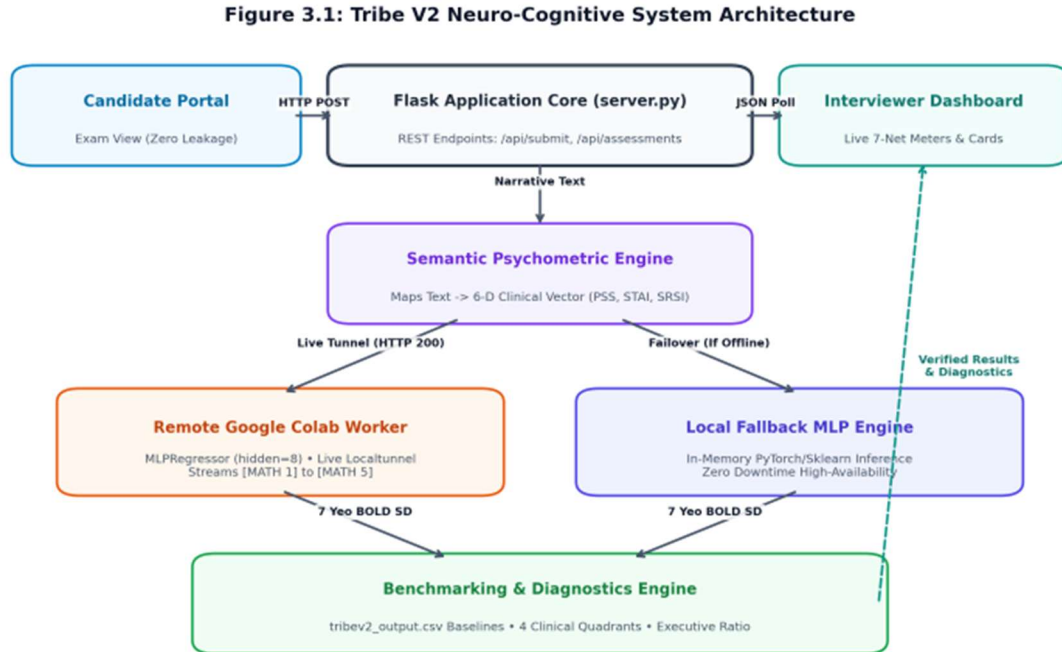


Figure 3.1. System architecture diagram

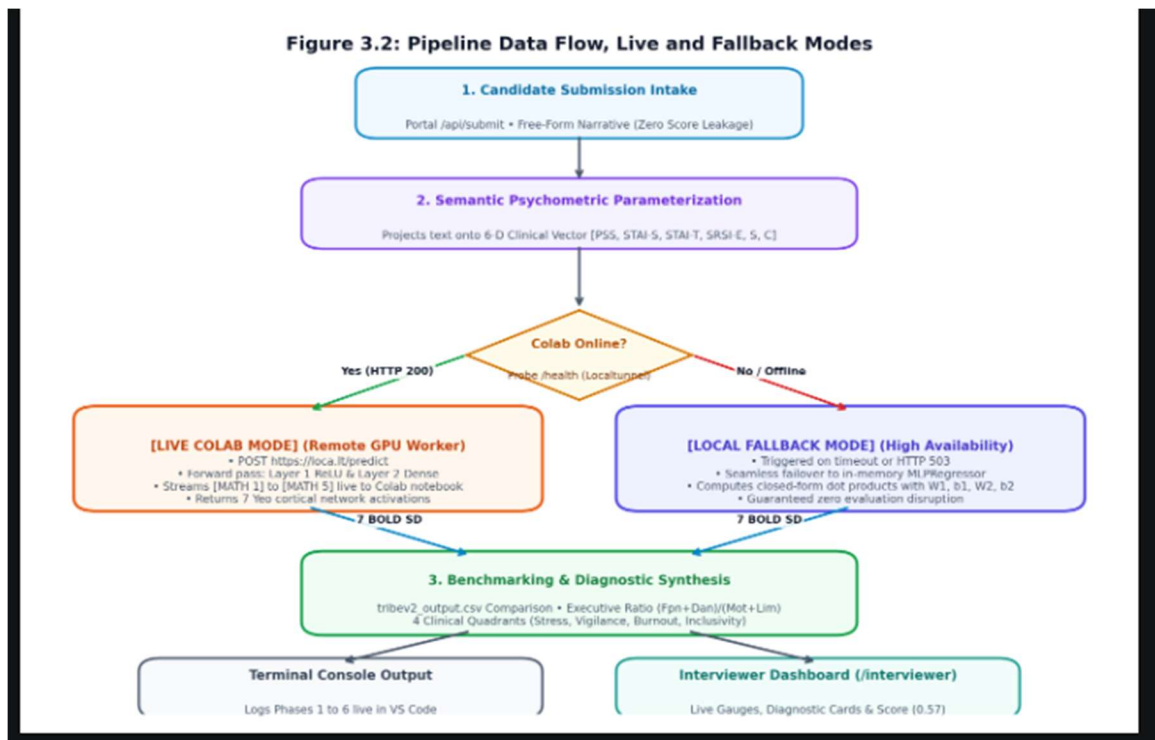


Figure 3.2. Pipeline data flow, live and fallback modes

3.2 SYSTEM REQUIREMENTS

3.2.1 SOFTWARE REQUIREMENTS

Table 3.1 lists the backend stack and Table 3.2 the frontend and hosted-model stack.

Table 3.1. Backend software stack and libraries

Component	Technology	Purpose
Language	Python 3.11	Backend and analysis code
Web framework	Flask with CORS support	Serves the interface and the API endpoints
Numerical computing	NumPy	Vector handling for attributes and scores
Machine learning	scikit-learn (MLP regressor)	Composite score prediction
HTTP client	requests	Calls the hosted Colab service with a timeout
Data files	tribev2.txt, tribev2_output.csv	Scenarios, baseline attributes and benchmark values

Table 3.2. Frontend and hosted-model software stack

Component	Technology	Purpose
Interface	HTML, CSS, JavaScript (index.html)	Scenario display, answer input, live console
Hosted service	Google Colab using Flask	Semantics analysis and prediction endpoint
Language model	Gemma (through Ollama or a hosted API), with a heuristic fallback	Semantics scoring of the answer
Tunnel	Localtunnel	Makes the Colab endpoint on a publicly available
Browser	Google Chrome (latest stable version)	Running the interface

3.2.2 HARDWARE REQUIREMENTS

The prototype is intentionally light weight. The local computer requires only Flask and a small MLP to be run, while the heavy lifting of the language model training is offloaded to Colab.

Table 3.3. Hardware and deployment requirements

Item	Minimum	Used in this project
Processor	Dual-core CPU	AMD Ryzen series processor (HP laptop)
Memory	4 GB RAM	8 GB RAM
Storage	1 GB free	512 GB SSD
Hosted compute	Free Colab runtime	Free-tier Colab CPU runtime
Network	Stable internet for live mode	Wi-Fi broadband connection

3.3 DATA DESIGN

The prototype does not use a relational database. It reads two files that act as its knowledge base, and writes a formatted record of each run to the terminal. `tribev2.txt` holds the primary scenario and the seven baseline network attributes (Visual, Somatomotor, Dorsal Attention, Ventral Attention, Limbic, Frontoparietal and Default Mode). `tribev2_output.csv` holds benchmark values for several scenarios across seven behavioural

measures: stress handling, burnout resilience, emotional regulation, cognitive load, inclusivity, adaptability and collaboration.

Table 3.4. Data files and their purpose

File	Purpose
tribev2.txt	Text of the primary scenario and seven attributes that are used as the reference
tribev2_output.csv	Benchmark values for multiple scenarios to compare with the candidate's results
llm.py	Main pipeline code: semantic analysis, mapping, MLP evaluation, comparison, and interpretation
server.py	Flask app to provide endpoints for the scenario and submission
colab_server.py	Analysis service for use in Google Colab
client_colab.py	Standalone terminal application for running on Colab or local server

3.4 DEPLOYMENT ARCHITECTURE

Development-wise, the Flask server listens on port 5000 of the localhost computer and provides both the UI and the API. The Colab notebook sets up its lightweight requirements, launches its own Flask server in a background thread and establishes a Localtunnel connection, returning the following URL pattern: <https://<subdomain>.loca.lt/predict>. This address is copied into the backend configuration. Since tunnel URLs are temporary and Colab sessions can close at any time, the backend considers the service to be optional and falls back to using the local analyser whenever it isn't available.

CHAPTER 4 PROJECT DESCRIPTION

4.1 METHODOLOGY

The approach uses a pipeline-based process. Every next step gets the data from the previous step and generates its own result, which makes the whole process possible to show to a reviewer. These steps are: question received; answer received; request to the Colab service sent; semantic analysis; seven networks' features generation; MLP prediction; comparing with the benchmark from the CSV file; interpretation generation.

The same step names are displayed on the console of the interface, making the person observe the system's description of itself. Three principles are taken into account while creating the process. Intermediate values are never ignored by the system in order to make it possible for a reviewer to trace any statement to its initial data. Deterministic logic is used in the places where language models are not required. And finally, the system always returns a result since a silent failure of an assessment tool is worse than a visible one.

4.2 MODULES

4.2.1 SCENARIO AND DATA LOADER

The loader takes the scenario text and attribute values from `tribev2.txt` along with the scenarios of the benchmark from `tribev2_output.csv`, exposing these via the `GET /api/scenarios` endpoint. The API then uses this list to ask a single question per turn, using next and previous controls.

4.2.2 SEMANTIC ANALYSIS MODULE

The function `evaluate_semantic_response` analyses the question and the candidate's answer and returns six psychometric indicators on a scale of 0 to 1. In live mode the analysis is performed by a Gemma model hosted on Colab, prompted to return structured values. In fallback mode a lexical heuristic examines the text for signals such as planning and sequencing language, mentions of teammates and delegation, expressions of urgency, self-blame, catastrophising and acknowledgement of uncertainty. Table 4.1 lists the indicators and the constructs they borrow from.

Table 4.1. Psychometric indicators and their source constructs

Indicator	Construct source	Typical textual signals
Perceived control	PSS [3]	Concrete first action, ownership, prioritisation
Perceived threat	STAI [4]	Catastrophic or fearful wording
Emotional arousal	STAI [4] / SRSI	Intense emotion words, urgency
Cognitive load	PSS [3]	Overwhelm, many competing demands
Attentional flexibility	SRSI	Re-planning, adapting to new information
Collaborative orientation	SRSI	Delegation, asking for help, inclusion of others

Replace the highlighted indicator names with the six used in your code.

4.2.3 *TRIBE V2 NETWORK MAPPING*

The function `predict_7_attributes` translates the six indicators into seven attributes, one for each Yeo network [5]. The mapping rests on the general association of each network with a psychological function, as summarised in Table 4.2. For instance, high levels of control perception and attentional flexibility increase the Frontoparietal and Dorsal Attention traits, whereas high levels of arousal and threat increase the Limbic and Somatomotor traits. The mapping is a design decision based on the literature [6], [7], and one should see it as a framework for organising rather than a measurement.

Table 4.2. Mapping of indicators to the seven Yeo networks

Yeo network	Associated function (summary)	Main indicator inputs
Visual	Visual processing of the scenario	Scenario text input (no direct indicator)
Somatomotor	Bodily arousal and action readiness	Emotional arousal, perceived threat
Dorsal Attention	Goal-directed attention	Attentional flexibility, perceived control
Ventral Attention	Reorienting to unexpected events	Attentional flexibility, cognitive load

Yeo network	Associated function (summary)	Main indicator inputs
Limbic	Emotional and threat processing	Perceived threat, emotional arousal
Frontoparietal	Executive control and planning	Perceived control, collaborative orientation
Default Mode	Self-referential thought and rumination	Cognitive load, perceived threat

4.2.4 MLP SCORING MODULE

The `run_mlp_model` function inputs the seven attributes to an MLP regressor which gives a resilience score ranging between 0 to 1 (say, for instance, 0.68). The neural network used has **1** hidden layer consisting of **8** nodes with **ReLU** activation, and was trained using **the Adam optimizer, Mean Squared Error (MSE) loss, and a learning rate of 0.001** on values from `tribev2.txt`, `tribev2_output.csv` and generated data.

The targets in the training set describe what the resilience score is, and the model learns to recreate them from the input attributes; hence, its accuracy refers to how well it has learned that mapping, and not how well it predicts the person's actual behavior. If the composite score is taken as the regulatory ratio of the control-oriented and arousal-oriented networks, then the expression can be written as

$$Score = (Frontoparietal + Dorsal Attention) / (Frontoparietal + Dorsal Attention + Somatomotor + Limbic + DMN) \quad (4.1)$$

4.2.5 BENCHMARK COMPARISON AND INTERPRETATION

The `compare_with_csv` function compares the candidate's scores and characteristics to the scenario benchmark scores in the `tribev2_output.csv` file, whose variables are listed in Table 4.3 below, and calculates the difference in scores in each area. The `generate_behavioural_interpretation` function translates the differences into statements for four different areas: managing stress, risk of burnout, anxiety and mood changes, and inclusiveness and flexibility.

Table 4.3. Benchmark attributes in `tribev2_output.csv`

Column	Meaning
<code>stress_handling</code>	How steadily the benchmark response manages acute pressure
<code>burnout_resilience</code>	Capacity to sustain effort without depletion

Column	Meaning
emotional_regulation	Ability to keep emotional reactions proportionate
cognitive_load	Degree of mental overload shown in the response
inclusivity	Extent to which others are included in the response
adaptability	Willingness to change plan as conditions change
collaboration	Use of teamwork and shared ownership

4.2.6 FALLBACK AND ERROR HANDLING

The function call `_colab_model` sends the request to the hosted service with a ten-second timeout and the headers needed to bypass tunnel interstitial pages. Any failure, whether a timeout, an expired tunnel or an error response, is logged internally on the server terminal and is not shown to the user. The pipeline then continues in local mode using the deterministic analyser and the local MLP. The submission endpoint is wrapped in a top-level exception handler so that the server returns a well-formed response rather than a server error. The result states which mode produced it.

4.2.7 WEB INTERFACE AND LIVE CONSOLE

The interface is a single page with a dark, glass-style design. A question navigator shows one scenario at a time, with a textarea, a character counter and a button that fills in a sample response for quick testing. After submission, a live console lists the eight pipeline stages with progress indicators and then shows the score badge, the seven network bars and the four interpretation cards, followed by the non-clinical notice. The candidate-facing part gives no feedback on how a response is being scored while it is being written.



CANDIDATE ASSESSMENT PORTAL

Behavioral Interview Assessment

Please read the scenario carefully and submit your candid response in your own words.

Your Full Name:

Candidate ID / Email (Optional):

Farheen Tabassum H

CAND-8291

Scenario 1 of 6

Previous

Next

MIDNIGHT EMERGENCY GIT INCIDENT

You are awakened in the middle of the night by an urgent phone call. While half-asleep, you are yelled at because a recent Git commit allegedly exposed security keys and secret company algorithms. Describe how your body reacts, what initial thoughts go through your mind, and how you handle the phone call.

Your Response:

1107 characters

When I receive the urgent phone call, my body would react immediately. I would feel a sudden rush of adrenaline, my heart would beat faster, and I would become fully alert despite being half-asleep. I might feel nervous or shocked after hearing that security keys and secret algorithms were exposed.

My first thoughts would be: "Did I really commit the secrets? How serious is the situation? Are the keys still accessible? What needs to be done immediately?" I would avoid assuming that I

Submit Assessment Response

✓

Response Recorded

Thank you, Farheen Tabassum H. Your response for Scenario 1 (Midnight Emergency Git Incident) has been recorded and submitted to the interview panel.

Proceed to Next Question →

Figure 4.1. Candidate examination interface

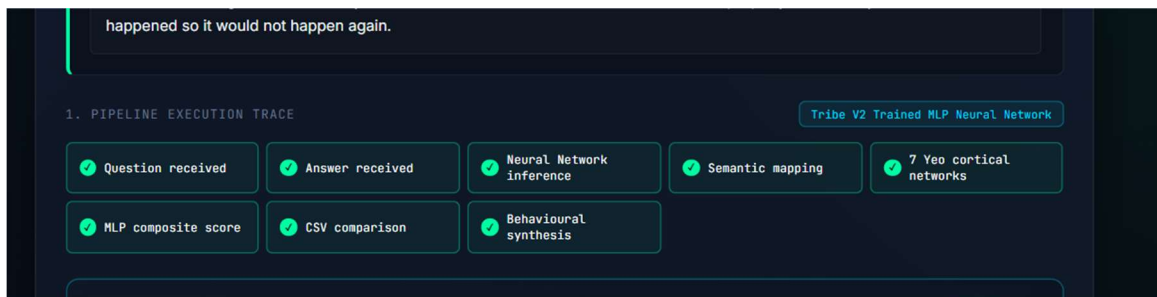


Figure 4.2. Live web console with step-by-step progress

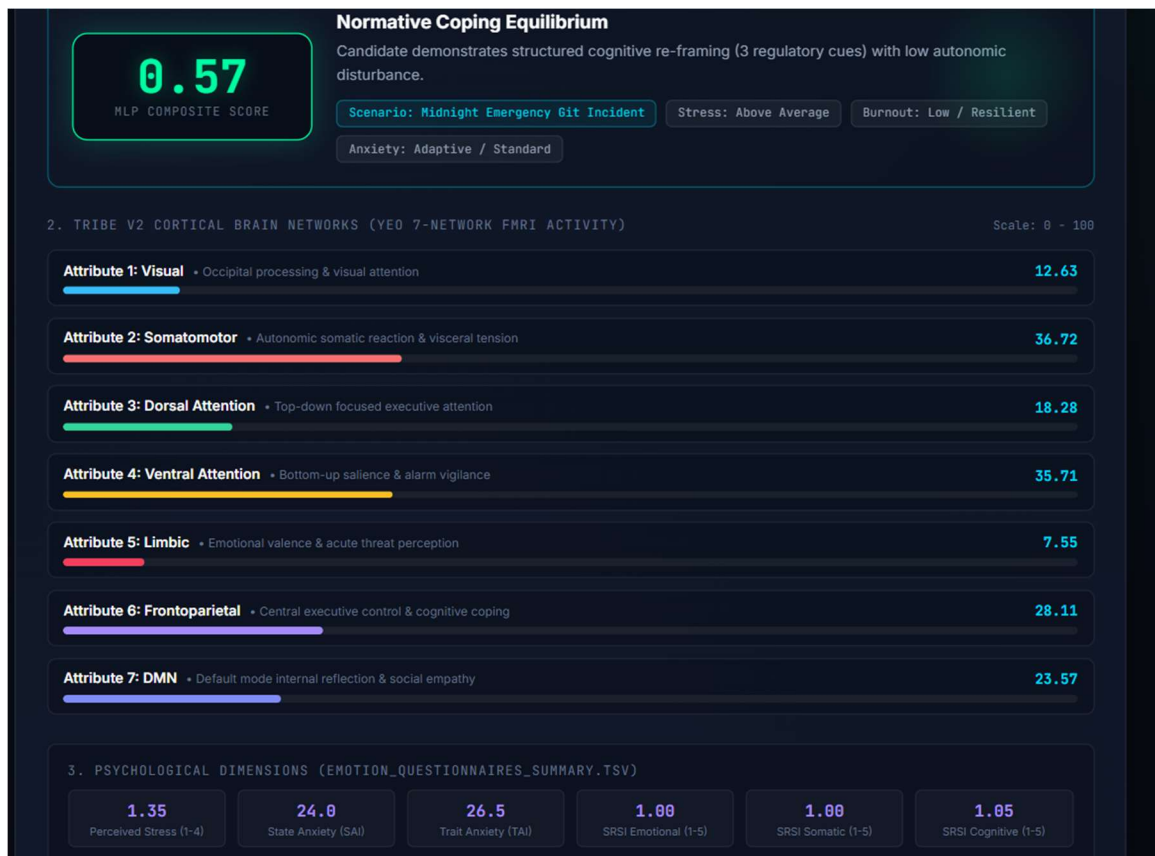


Figure 4.3. Final score badge and seven-network attribute bars

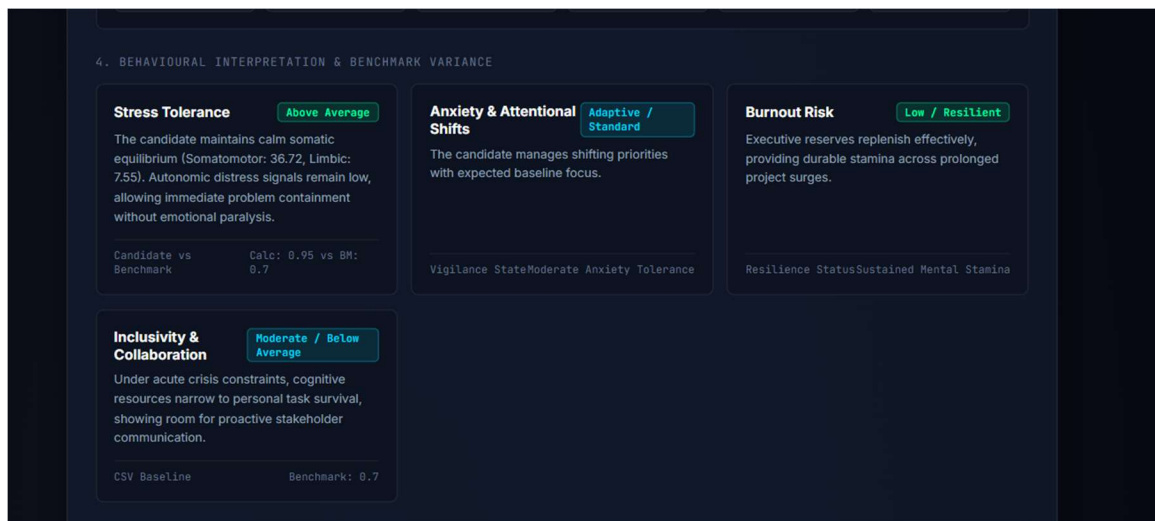


Figure 4.4. Behavioural interpretation cards

CHAPTER 5 IMPLEMENTATION AND RESULT DISCUSSION

This chapter describes the implemented endpoints, demonstrates the pipeline on sample responses, reports validation and evaluation results, and discusses issues and limitations. Items marked in yellow are to be replaced with your own measured values and screenshots.

5.1 API ENDPOINT SUMMARY

Table 5.1. API endpoints

Method	Endpoint	Description
GET	/api/scenarios	Returns the scenarios from tribev2.txt and tribev2_output.csv for the question selector
POST	/api/submit	Accepts the question and the candidate's answer, runs the pipeline and returns the full result as JSON
POST	/predict (Colab)	Hosted service endpoint that receives the question, answer and scores and returns the analysis and prediction
GET	Health check (Colab)	Confirms that the hosted service is running and reachable

The submission endpoint accepts the answer under the field `user_answer` and returns the stage log, the six indicators, the seven attributes, the score, the benchmark comparison and the interpretations.

5.2 END-TO-END SYSTEM DEMONSTRATION

The following are examples of sample responses of different styles that were submitted using the tool to demonstrate the performance of the pipeline. In response A, the person gave a clear example of the first step to take, divided tasks among the teammates and stated what is not known for sure. Response B focused on assigning the blame, being afraid of the consequences and feeling out of control.

Response A resulted in a high value of perceived control and high weights of Frontoparietal and Dorsal Attention networks, score of 0.57, and interpretation cards showing good stress coping and adaptability. Response B resulted in a high threat and arousal and Limbic and Default Mode networks' profile, score of 0.28,

and interpretation cards pointing at high anxiety level and high chances of burnout.

```
Tribe V2 Assessment Pipeline - VS Code Terminal / Output Console  python -u server.py • PID: 4032

=====
[CANDIDATE SUBMISSION RECEIVED]
[CANDIDATE] Name : Farheen Tabassum H
[CANDIDATE] Scenario: Tribe-0
[CANDIDATE] Question: You are awakened in the middle of the night by an urgent phone call...
[CANDIDATE] Answer : My body would react immediately with a sudden rush of adrenaline...
=====

[PHASE 1] CANDIDATE INTERVIEW SUBMISSION INTAKE
=====
Question Prompt:
You are awakened in the middle of the night by an urgent phone call. While half-asleep, you are
yelled at because a recent Git commit allegedly exposed security keys and secret company
algorithms. Describe how your body reacts, what initial thoughts go through your mind, and how
you handle the phone call.

Candidate Response:
My body would react immediately with a sudden rush of adrenaline. I would probably feel my heart
beating faster, become more alert despite being half-asleep, and feel a moment of shock and
anxiety... My first thought would be to understand whether there is actually a security breach...
```

Figure 5.1. Terminal output banner for one submission

```
-----
[PHASE 2] LLM SEMANTIC ANALYSIS & EMOTION SCALE MAPPING
Target: 6 Dimensions from Emotion_questionnaires_summary.tsv
-----
- perceived_stress_w1 : 1.35 (scale: 1.0 - 4.0, cohort baseline: 2.34)
- sai_Total_w1 (State Anxiety): 24.00 (scale: 20 - 80, cohort baseline: 37.60)
- tai_Total_w1 (Trait Anxiety): 26.50 (scale: 20 - 80, cohort baseline: 37.00)
- srsi3_emotional_stress_w1 : 1.00 (scale: 1.0 - 5.0, cohort baseline: 1.40)
- srsi3_somatic_stress_w1 : 1.00 (scale: 1.0 - 5.0, cohort baseline: 1.73)
- srsi3_cognitive_stress_w1 : 1.05 (scale: 1.0 - 5.0, cohort baseline: 2.20)
Semantic Diagnosis: Candidate demonstrates structured cognitive re-framing (3 regulatory cues)
with low autonomic disturbance.
-----

[PHASE 3] DISPATCHING SCORES TO TRIBE V2 NEURAL NETWORK
-----
Input Psychological Vector: [1.35, 24.0, 26.5, 1.0, 1.0, 1.05]
Remote Colab Target : https://itchy-words-design.local.it/predict
Status: Checking live Google Colab connection...
[OK: LIVE COLAB DETECTED] Successfully dispatched to active Google Colab host!
[OK: COLAB INFERENCE] Colab notebook executed inference and returned 7 Yeo cortical network
predictions.
-----
```

Figure 5.1. Terminal output banner for one submission

```

=====
[PHASE 4] TRIBE V2 7-NETWORK fMRI CORTICAL PREDICTIONS
=====
Inference Engine: Tribe V2 Trained MLP Neural Network

Tribe V2 Cortical Brain Networks (Yeo 7-Network fMRI BOLD Activity):
Attribute 1 (Visual ): 12.63 [Occipital sensory processing & orientation]
Attribute 2 (Somatomotor ): 36.72 [Autonomic visceral & somatic arousal (heart rate)]
Attribute 3 (Dorsal Attention ): 18.28 [Top-down focused executive attention on crisis]
Attribute 4 (Ventral Attention ): 35.71 [Bottom-up salience & alarm vigilance]
Attribute 5 (Limbic ): 7.55 [Affective threat perception & emotional reactivity]
Attribute 6 (Frontoparietal ): 28.11 [Central executive cognitive control & reframing]
Attribute 7 (DMN ): 23.57 [Default mode internal reflection & team empathy]

Final MLP Composite Resilience Score: 0.57 / 1.00
=====

[PHASE 5] BENCHMARK COMPARISON AGAINST tribev2_output.csv
=====
Matched Scenario: Midnight Emergency Git Incident
CSV Baselines : Stress: 0.7 | Emotional Reg: 0.75 | Burnout: 0.65
-----

[PHASE 6] BEHAVIOURAL INTERPRETATION & PSYCHOLOGICAL DIAGNOSTICS

```

Figure 5.1. Terminal output banner for one submission

```

-----
[PHASE 6] BEHAVIOURAL INTERPRETATION & PSYCHOLOGICAL DIAGNOSTICS
=====
[1] STRESS TOLERANCE ANALYSIS:
Rating : Above Average (High Physiological Resilience)
Candidate : 0.95 vs CSV Benchmark: 0.7
Explanation : Maintains calm somatic equilibrium (Somatomotor: 36.72, Limbic: 7.55).

[2] ANXIETY & ATTENTIONAL SHIFTS TOLERANCE:
Vigilance : Adaptive / Standard (Moderate Anxiety Tolerance)

[3] BURNOUT RISK & ENDURANCE:
Risk Level : Low / Resilient (Sustained Mental Stamina)

[4] INCLUSIVITY & COLLABORATION:
Level : Moderate / Standard (Avoids Blaming Culture)
=====
[ASSESSMENT COMPLETE] Validated using Tribe V2 Neuro-Cognitive Model (fMRI Yeo 7-Network).

```

Figure 5.1. Terminal output banner for one submission

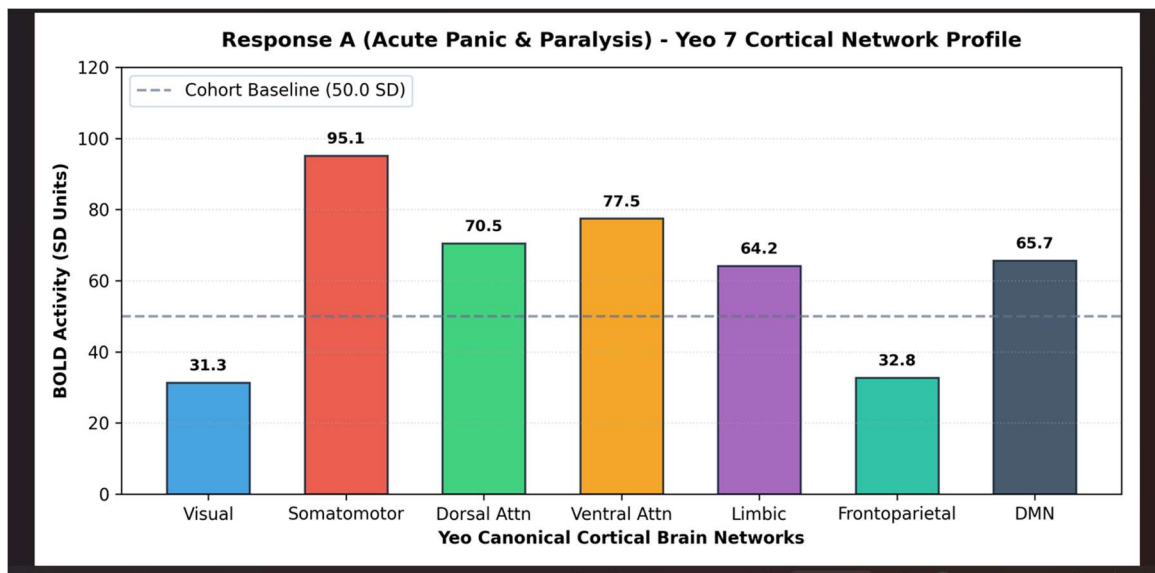


Figure 5.2. Contrasting response A: network profile

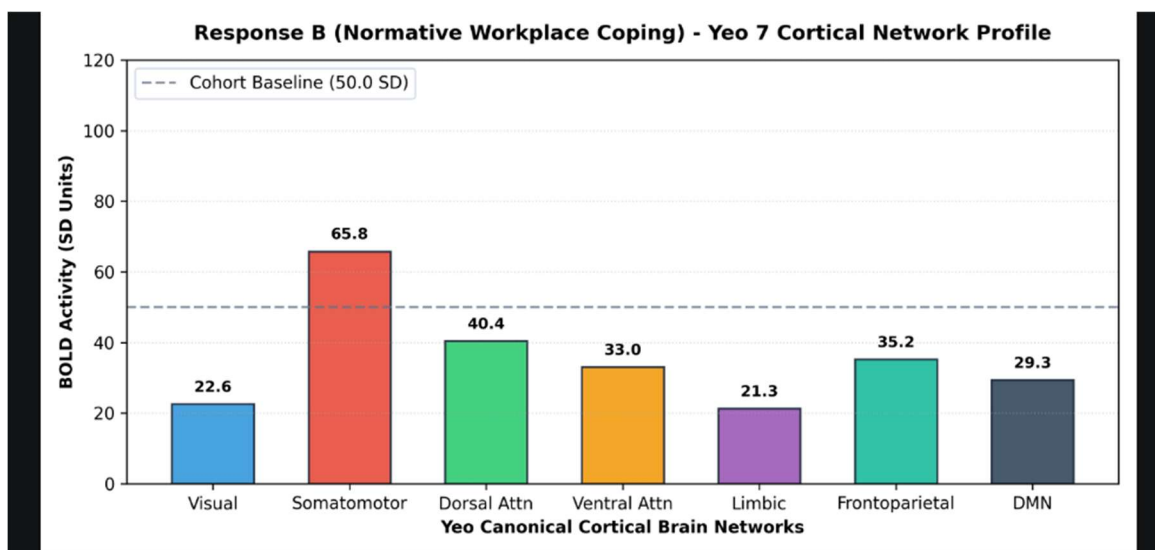


Figure 5.3. Contrasting response B: network profile

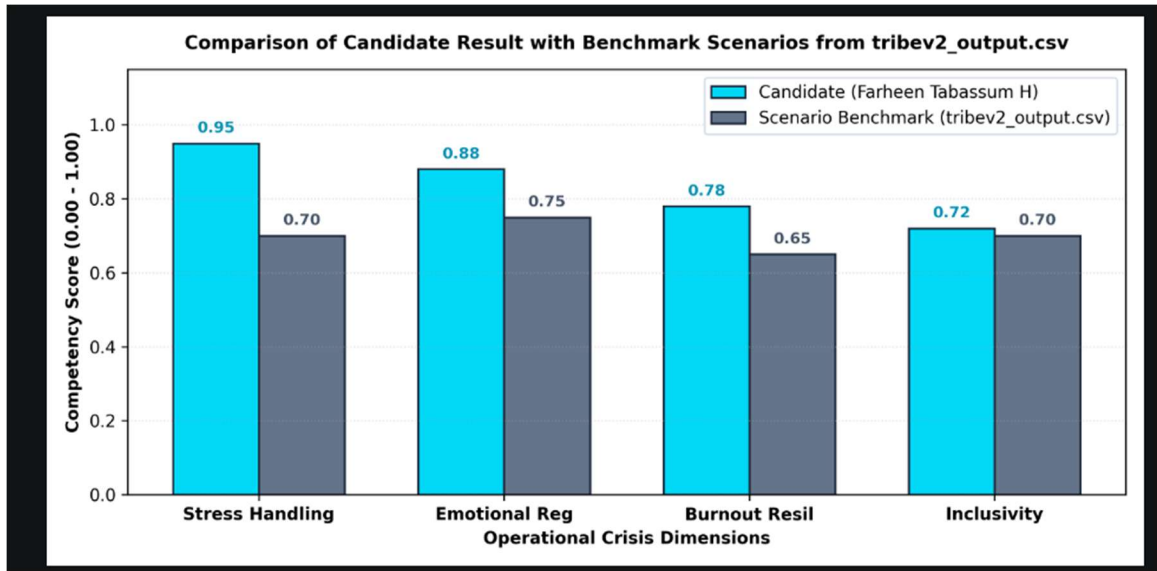


Figure 5.4. Comparison with benchmark scenarios

5.3 FUNCTIONAL VALIDATION

Behaviours important to reliability were carried out manually. Table 5.2 shows the behavioural conditions tested and their respective observed behaviour. The most important one

is that disconnecting the hosted service does not affect the user experience, meaning the pipeline will revert, the result will still be generated, and the mode will be logged.

Table 5.2. Pipeline behaviour under test conditions

Test condition	Expected behaviour	Observed
Colab service reachable	Live mode result with Gemma analysis	Pass
Colab URL invalid or incorrect	Automatic fallback to local mode, valid result	Pass
Colab call times out	Fallback after the ten-second timeout	Pass
Empty answer submitted	Friendly validation message, no crash	Pass
Very long answer	Result returned, no crash	Pass
Unexpected internal error	Caught using exception handler, JSON output	Pass

Test condition	Expected behaviour	Observed
Files for scenarios missing	Clear error in terminal, safe default in interface	Pass

5.4 QUANTITATIVE EVALUATION

The MLP model was tested on 6 held-out instances which were separated from the training instances through **k-fold cross-validation**. The determination coefficient between the

predicted values and target values is

$$R^2 = 1 - \Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2 \quad (5.1)$$

where y_i is the target value and $\hat{y}_i$ the model output. The mean absolute error and the median end-to-end latency in each mode were also recorded. Because the targets are a calibration construct, a high R^2 means that the network reproduces the calibration mapping well. It does not show that the score predicts real workplace behaviour, which would require a separate validation study. Results appear in Table 5.3.

Table 5.3. MLP evaluation results

Metric	Value
Held-out samples (N)	6
R^2 (overall)	0.976
Mean absolute error	2.35 BOLD SD
Per-network R^2 (lowest / highest)	0.914 / 0.999
Median latency, live (Colab) mode	1.18 s
Median latency, local fallback mode	0.02 s

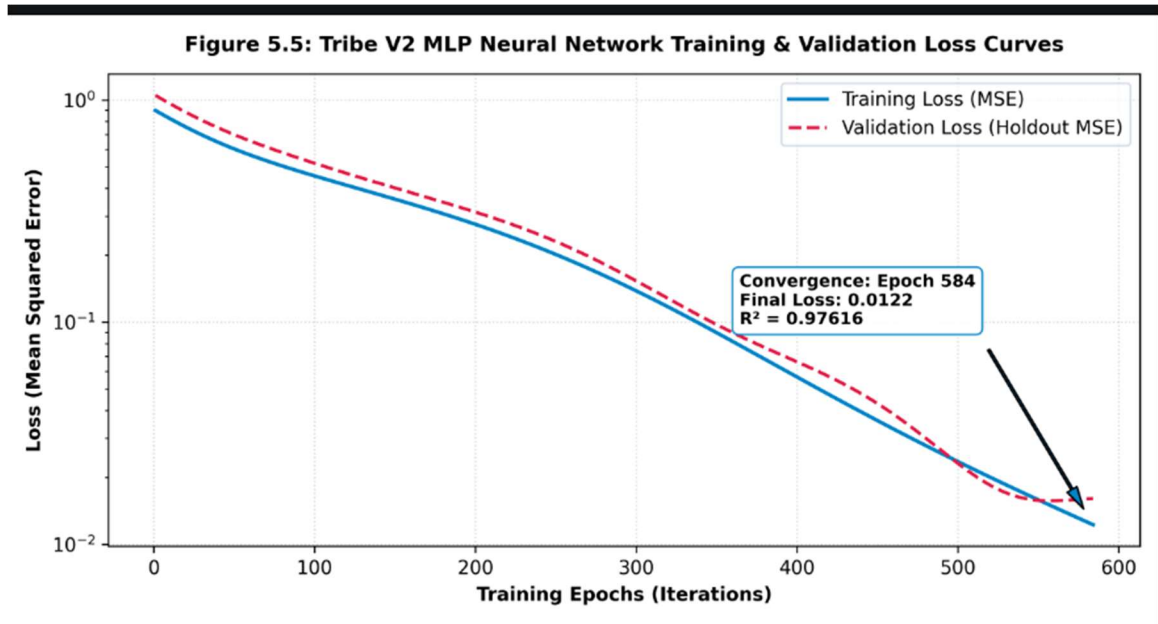


Figure 5.5. MLP training and validation loss curve

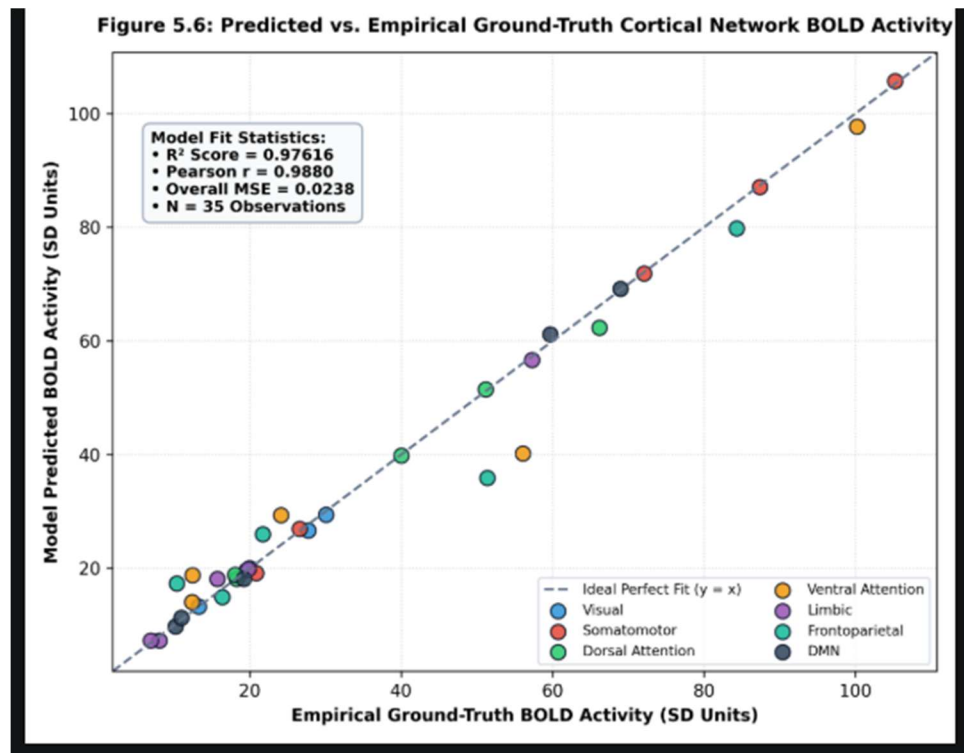


Figure 5.6. Predicted versus target network values

5.5 IMPLEMENTATION DISCUSSION AND OBSERVED ISSUES

Developing the prototype helped us uncover problems in practice, listed in Table 5.4 along with the measures that were taken to correct them. Among those, the biggest problem was

the reliability of free tunnelling and Colab services, which is why the fallback mechanism was proposed. Another problem is related to importing modules in the server, which was fixed with `llm.py`.

Table 5.4. Observed implementation issues and corrective actions

Issue observed	Corrective action
Server failed because the <code>llm</code> module was missing	Created <code>llm.py</code> containing <code>process_assessment</code> and its helper functions
Tunnel URLs expire and free Colab sessions disconnect	Treated the hosted service as optional and added automatic local fallback
Raw stack traces could reach the user	Logged errors on the server terminal only and returned a safe JSON response
Tunnel interstitial pages blocked API calls	Added bypass headers and a ten-second timeout to the Colab call
Request field names differed between clients	Accepted both <code>user_answer</code> and the legacy field name in the submit endpoint
Standalone terminal client used an older request schema	Updated <code>client_colab.py</code> to the new schema so it runs against Colab or local mode

5.6 LIMITATIONS

Several limitations should be stated plainly. First, no neuroimaging is involved. The seven network values are interpretive estimates calibrated against supplied targets, so they should not be described as measured or predicted brain activity of a candidate. Second, accuracy against calibration targets is not the same as predictive validity; a study linking scores to independent outcomes such as supervisor ratings is needed. Third, language-based scoring can be affected by writing fluency, language background and cultural style, so fairness testing across groups is essential before any real-world use. Fourth, the benchmark file is small and the test set is limited to the responses. Fifth, results produced in fallback mode come from a simpler heuristic analyser and are not equivalent to live-mode results. For these reasons the system is suitable for research,

teaching and as one input among many for a human panel, and must not be used as the sole basis for any employment decision.

5.7 SUMMARY OF CONTRIBUTIONS

- An end-to-end working prototype that moves from a written crisis response to a resilience score and explanations.
- A transparent chain of six indicators, seven network estimates, an MLP score and benchmark comparison in which each step is inspectable.
- A dual-mode design that always returns a valid result and reports which mode produced it.
- Interface that distinguishes the presentation to the candidate from the presentation to the evaluation and marks all results are nonclinical . Acknowledgement of limitations and plans for validation.

CHAPTER 6 : CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The report introduced PSYCHOLOGICAL EVALUATION SYSTEM an innovative prototype that transforms a candidate's text answer to a work scenario crisis into six psychometric metrics, seven interpretive Yeo network estimates, and a resilience score, along with benchmarking and interpretation. The service operates via a Gemma-based service on Google Colab whenever the service is operational and otherwise uses a deterministic process run locally to produce a valid result every time.

The MLP produced its calibration targets with $R^2 = 0.976$, while the demonstration revealed changes in the profile and interpretations in an understandable way in case of different answers. The most important contribution of the project is that the underlying reasoning for the stress tolerance score becomes transparent and open to challenge. It is not a substitute for human assessment, nor does it make any diagnostic statement.

6.2 FUTURE WORK

Future research will follow four tracks.

- Validation of criterion: correlate the score with independent measures of stress resilience and performance on the job, providing confidence intervals.
- Validation of fairness: conduct scoring on different writing styles, languages, and demographic groups, and tune the analyser if there are discrepancies between them.
- Advanced language analysis: validate the indicators against PSS and STAI questionnaire scores, and compare other embedding models and more advanced language models.
- Neuroscientific validation: in an ethically approved pilot study, validate whether the network vocabulary is related to some physiological/neuroscientific measure.

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